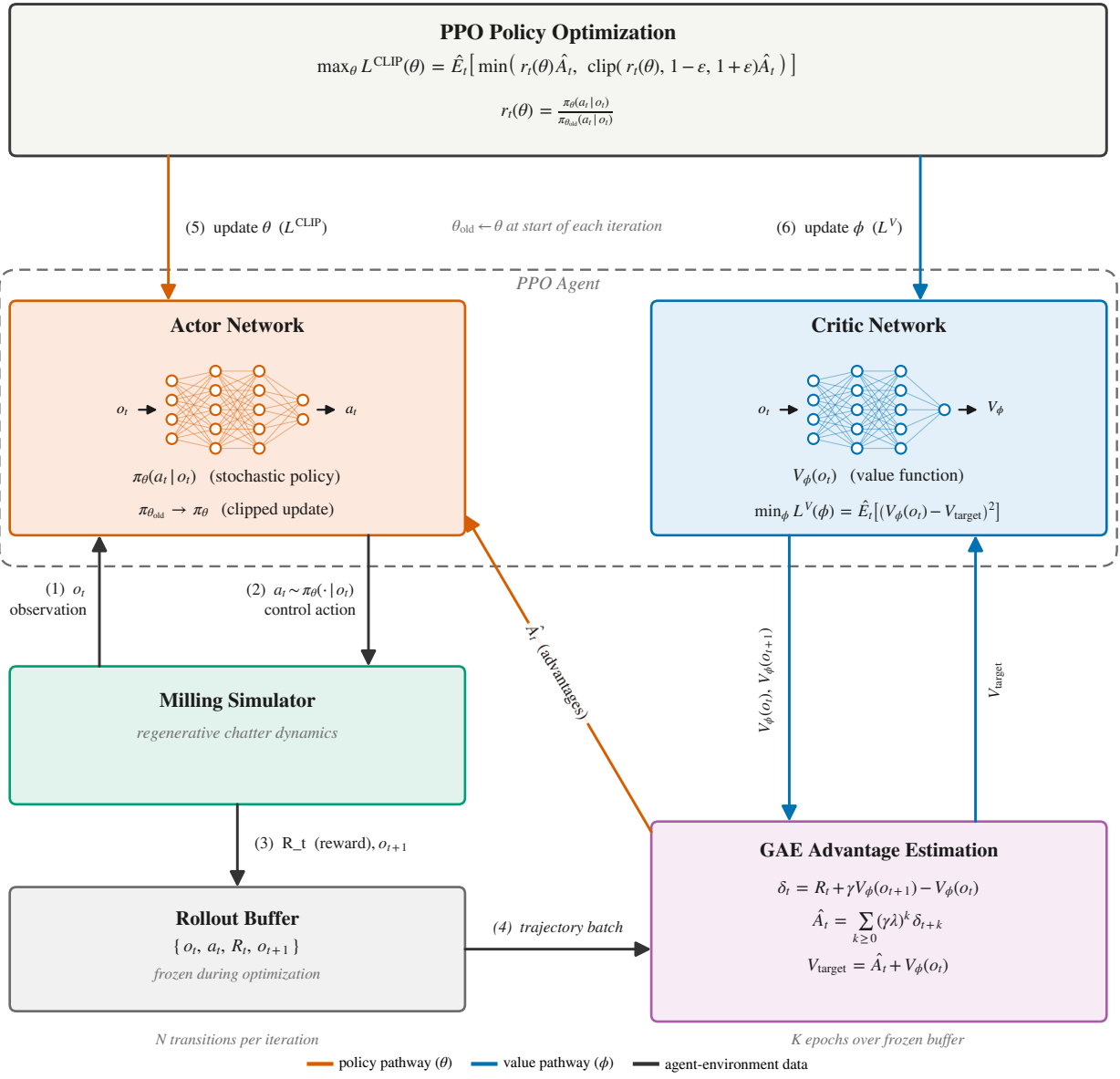
PPO Policy Optimization
$\max_\theta L^{\mathrm{CLIP}}(\theta) = \hat{E}_t\left[\min\left(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t\right)\right]$
$r_t(\theta) = \frac{\pi_\theta(a_t \mid o_t)}{\pi_{\theta_{\mathrm{old}}}(a_t \mid o_t)}$
(5) update θ (L^{CLIP})
$\theta_{\mathrm{old}} \leftarrow \theta$ at start of each iteration
(6) update ϕ (L^V)
PPO Agent
Actor Network
o_t
a_t
$\pi_\theta(a_t \mid o_t)$ (stochastic policy)
$\pi_{\theta_{\mathrm{old}}} \rightarrow \pi_\theta$ (clipped update)
Critic Network
o_t
V_ϕ
$V_\phi(o_t)$ (value function)
$\min_\phi L^V(\phi) = \hat{E}_t[(V_\phi(o_t) - V_{\mathrm{target}})^2]$
(1) o_t observation
(2) $a_t \sim \pi_\theta(\cdot \mid o_t)$ control action
$\hat{A}_t$ (advantages)
$V_\phi(o_t)$, $V_\phi(o_{t+1})$
V_{target}
Milling Simulator
regenerative chatter dynamics
(3) R_t (reward), o_{t+1}
GAE Advantage Estimation
$\delta_t = R_t + \gamma V_\phi(o_{t+1}) - V_\phi(o_t)$
$\hat{A}_t = \sum_{k \geq 0} (\gamma\lambda)^k \delta_{t+k}$
$V_{\mathrm{target}} = \hat{A}_t + V_\phi(o_t)$
Rollout Buffer
$\{o_t, a_t, R_t, o_{t+1}\}$
frozen during optimization
(4) trajectory batch
N transitions per iteration
K epochs over frozen buffer
policy pathway (θ)
value pathway (ϕ)
agent-environment data